CHAPTER NOTES

DESCRIBING MOTION AROUND US

CLASS 9 • PHYSICS

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• Distance is a **scalar** quantity: it has magnitude only.

• It is always zero or positive; it cannot be negative.

• SI unit: metre (m). Other common units include kilometre (km) and centimetre (cm).

• Distance depends on the actual path followed.

Displacement

Displacement is the shortest straight-line change in position from the initial point to the final point, together with direction. It is represented by a directed line from start to finish.

• Displacement is a **vector** quantity: it has magnitude and direction.

• Its magnitude is never greater than the distance travelled.

• It can be positive, negative or zero in one-dimensional motion, depending on the chosen direction.

• For a round trip ending at the starting point, displacement is zero.

Feature	Distance	Displacement
Meaning	Total path length	Shortest straight-line change from start to finish
Type	Scalar	Vector
Direction	Not required	Required
Path	Depends on actual path	Depends only on initial and final positions
Value	Non-negative	May be positive, negative or zero (in 1-D)

Worked examples

Example 1: A student walks 3 m east and then 4 m west. Total distance = 3 + 4 = **7 m**. Taking east as positive, displacement = 3 – 4 = –**1 m**, i.e. 1 m west.

Example 2: A runner completes one full lap of a 400 m track and returns to the start. Distance = **400 m**; displacement = **0 m**.

Key relation: Magnitude of displacement ≤ distance. Equality occurs when an object moves along a straight line without changing direction.

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4. SCALARS AND VECTORS

Physical quantities used to describe motion are often grouped by whether direction is needed.

Scalar quantities	Vector quantities
Distance	Displacement
Speed	Velocity
Time	Acceleration
Mass	Force (studied in detail later)

How to describe a vector

A vector is written with a magnitude and a direction, for example 5 m east or 12 m/s north. A vector can be represented by an arrow: its length indicates magnitude and its arrowhead indicates direction.

Choosing a positive direction

In straight-line motion, we often choose one direction as positive. The opposite direction is then negative. If east is positive, west is negative. The sign of displacement, velocity or acceleration depends on this choice; the physical direction does not change.

Illustration in words: On a straight road, mark the starting point as O. Positions to the right are positive and positions to the left are negative. Moving from –2 m to +5 m gives displacement +7 m.

5. MEASURING TIME AND POSITION

SI units

Quantity	SI unit	Symbol
Length / distance / displacement	metre	m
Time	second	s
Speed / velocity	metre per second	m/s
Acceleration	metre per second squared	m/s²

Useful unit conversions

• 1 kilometre (km) = 1000 metres (m).

• 1 hour (h) = 3600 seconds (s).

• To convert km/h into m/s, multiply by 5/18.

• To convert m/s into km/h, multiply by 18/5.

Example: 72 km/h = 72 × 5/18 = **20 m/s**. And 15 m/s = 15 × 18/5 = **54 km/h**.

6. SPEED

Speed tells us how fast an object covers distance. It is the distance travelled per unit time.

Average speed = Total distance travelled ÷ Total time taken

SI unit of speed is metre per second (m/s). Speed is a scalar quantity because it does not specify direction.

Uniform and non-uniform speed

• **Uniform speed:** an object covers equal distances in equal intervals of time, however small the intervals may be.

• **Non-uniform speed:** an object covers unequal distances in equal intervals of time, or equal distances in unequal intervals of time.

Worked examples

Example 1: A car travels 150 km in 3 h. Average speed = 150 ÷ 3 = **50 km/h**.

Example 2: A cyclist covers 600 m in 2 minutes. Time = 120 s. Average speed = 600 ÷ 120 = **5 m/s**.

Example 3: A vehicle travels 40 km in the first hour and 60 km in the next hour. Total distance = 100 km; total time = 2 h; average speed = **50 km/h**. It is not found by simply averaging speeds unless the time intervals are equal.

7. VELOCITY

Velocity is the displacement per unit time. It describes both how fast an object moves and the direction of its motion.

Average velocity = Total displacement ÷ Total time taken

• Velocity is a vector quantity and has direction.

• SI unit: m/s.

• For straight-line motion, velocity is positive or negative according to the chosen positive direction.

• An object can have non-zero average speed but zero average velocity over a complete round trip.

Speed versus velocity

Speed	Velocity
Based on distance	Based on displacement
Scalar	Vector
No direction needed	Direction is essential
Cannot be negative	May be positive, negative or zero in 1-D

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8. ACCELERATION

Acceleration is the rate of change of velocity with time. An object accelerates if its velocity changes in magnitude, direction, or both.

Acceleration = Change in velocity ÷ Time taken
a = (v – u) / t

Here, u = initial velocity, v = final velocity, t = time interval and a = acceleration. SI unit: m/s².

Types of acceleration

• **Positive acceleration:** acceleration in the chosen positive direction.

• **Negative acceleration:** acceleration in the chosen negative direction. It is sometimes called retardation when it acts opposite to the velocity and reduces speed.

• **Uniform acceleration:** velocity changes by equal amounts in equal time intervals.

• **Non-uniform acceleration:** velocity changes by unequal amounts in equal time intervals.

Worked examples

Example 1: A scooter's velocity increases from 5 m/s to 15 m/s in 5 s. a = (15 – 5)/5 = **2 m/s²**.

Example 2: A car slows from 20 m/s to 8 m/s in 4 s. Taking forward as positive, a = (8 – 20)/4 = –**3 m/s²**. The negative sign indicates acceleration opposite to the chosen positive direction.

9. UNDERSTANDING MOTION THROUGH EVERYDAY EXAMPLES

Observation	What changes?	Motion description
A car moving faster	Speed increases	Accelerated motion
A bus braking	Velocity decreases	Negative acceleration if forward is positive
A turning bicycle	Direction changes	Velocity changes even if speed stays constant
A clock hand	Direction changes continuously	Circular motion

Important distinction: A change in direction alone is a change in velocity. Therefore, an object moving around a circular track at constant speed is still accelerating because its direction keeps changing.

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10. DISTANCE–TIME GRAPHS

A graph can show how an object's position or distance changes with time. In a distance–time graph, time is usually plotted on the horizontal (x) axis and distance on the vertical (y) axis.

Reading the shape of the graph

• A horizontal line means distance remains constant: the object is at rest.

• A straight sloping line means constant speed.

• A steeper line represents greater speed (for the same scale).

• A curved line indicates changing speed.

Finding speed from a distance–time graph

Speed = Slope of distance–time graph = Change in distance ÷ Change in time

For a straight line joining (t■, d■) and (t■, d■), slope = (d■ – d■)/(t■ – t■). Always include units in the answer.

Time (s)	Distance (m)	Interpretation
0	0	Starting point
2	10	10 m covered in 2 s
4	20	Another 10 m in 2 s
6	30	Uniform speed continues

For the table above, speed = 30 m ÷ 6 s = **5 m/s**. Equal increases in distance during equal time intervals indicate uniform speed.

11. DISPLACEMENT–TIME GRAPHS

A displacement–time graph uses displacement on the vertical axis and time on the horizontal axis. Its slope gives velocity. A positive slope indicates positive velocity, a negative slope indicates motion in the opposite direction, and a zero slope indicates rest.

Graph-reading checklist: Read the axis labels and units first; choose two points on the line; calculate change in vertical quantity divided by change in time; then state the correct unit and direction where relevant.

12. VELOCITY–TIME GRAPHS

A velocity–time graph shows how velocity changes with time. Time is on the x-axis and velocity on the y-axis.

Slope gives acceleration

Acceleration = Slope of velocity–time graph = Change in velocity ÷ Change in time

- A horizontal line means constant velocity and zero acceleration.
- A straight line sloping upward means uniform positive acceleration (with the chosen sign convention).
- A straight line sloping downward means uniform negative acceleration.
- A curved line indicates non-uniform acceleration.

Area under a velocity–time graph

For motion along a straight line, the signed area between the velocity–time graph and the time axis gives displacement. Area above the time axis is positive; area below it is negative.

Example: An object moves at a constant velocity of 4 m/s for 5 s. The graph is a horizontal line. Displacement = area of rectangle = 4 × 5 = **20 m**.

13. EQUATIONS OF UNIFORMLY ACCELERATED MOTION

For motion in a straight line with constant acceleration, the following equations are useful. They use a consistent sign convention.

First equation: $v = u + at$
Second equation: $s = ut + \frac{1}{2}at^2$
Third equation: $v^2 = u^2 + 2as$

Symbols: u = initial velocity; v = final velocity; a = constant acceleration; t = time; s = displacement. These equations should not be applied blindly when acceleration is changing.

Choosing the correct equation

If you know…	Useful equation
u, a, t; need v	$v = u + at$
u, a, t; need s	$s = ut + \frac{1}{2}at^2$
u, a, s; need v	$v^2 = u^2 + 2as$

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14. SOLVED NUMERICAL PROBLEMS

Use the given data, write the formula, substitute values with units, and state the final answer clearly.

Numerical 1 — finding final velocity

Question: A train starts with a velocity of 10 m/s and accelerates uniformly at 2 m/s² for 6 s. Find its final velocity.
Given: u = 10 m/s, a = 2 m/s², t = 6 s.
Formula: v = u + at.
Substitution: v = 10 + (2 × 6) = 22 m/s.
Answer: 22 m/s.

Numerical 2 — finding displacement

Question: A bicycle moving at 4 m/s accelerates at 1 m/s² for 5 s. Find displacement.
Given: u = 4 m/s, a = 1 m/s², t = 5 s.
Formula: s = ut + ½at².
Substitution: s = (4 × 5) + ½ × 1 × 25 = 20 + 12.5 = 32.5 m.
Answer: 32.5 m.

Numerical 3 — finding acceleration

Question: A car's velocity changes from 6 m/s to 18 m/s in 3 s. Find acceleration.
Formula: a = (v – u)/t = (18 – 6)/3 = **4 m/s²**.

Numerical 4 — average speed

Question: A person walks 120 m in 60 s and then 180 m in 90 s. Find average speed.
Total distance = 120 + 180 = 300 m.
Total time = 60 + 90 = 150 s.
Average speed = 300/150 = **2 m/s**.

15. COMMON ERRORS TO AVOID

- Do not use displacement when the question asks for total distance.
- Do not confuse speed (scalar) with velocity (vector).
- Convert hours into seconds when using SI units in a calculation.
- Keep the sign convention consistent throughout a numerical.
- Use total distance and total time for average speed; use total displacement and total time for average velocity.

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16. UNIFORM CIRCULAR MOTION

When an object moves along a circular path, its motion is called **circular motion**. If it covers equal distances along the circular path in equal intervals of time, it has uniform circular motion.

In uniform circular motion, the speed can remain constant, but velocity changes continuously because the direction of motion changes at every point on the circle. A change in velocity means acceleration is present.

Examples

• A stone tied to a string and rotated in a circle (under controlled conditions).

• The tip of the second hand of a clock.

• A satellite moving approximately in a circular orbit.

• A point on the rim of a rotating wheel.

Period and frequency — useful terms

Time period (T): time taken to complete one revolution. SI unit: second (s).
Frequency (f): number of revolutions completed per second. SI unit: hertz (Hz).
Relation: f = 1/T.

Example: If a wheel completes 2 revolutions in 1 second, its frequency is 2 Hz. Its time period is T = 1/2 = 0.5 s.

17. QUICK REVISION — FORMULA SHEET

Quantity / concept	Formula or key idea
Average speed	Total distance ÷ Total time
Average velocity	Total displacement ÷ Total time
Acceleration	a = (v – u)/t
First equation	v = u + at
Second equation	s = ut + ½at²
Third equation	v² = u² + 2as
Graph slope: distance–time	Speed
Graph slope: displacement–time	Velocity
Graph slope: velocity–time	Acceleration
Area under velocity–time graph	Displacement (signed area)
Frequency and period	f = 1/T

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18. PRACTICE QUESTIONS

Try these questions after revising the chapter. Show working for numerical problems and always write units.

A. Very short answer

• Define motion and name the quantity used as a reference when describing it.

• Is distance scalar or vector? What about displacement?

• Write the SI unit of acceleration.

• What does a horizontal line on a distance–time graph represent?

• Can an object have constant speed but changing velocity? Explain briefly.

B. Short answer

• Differentiate between distance and displacement with one example.

• Explain why rest and motion are relative.

• Distinguish between speed and velocity.

• What is uniform acceleration? Give one example.

• What information can be obtained from the slope of a velocity–time graph?

C. Numericals

• A student runs 100 m in 20 s. Calculate average speed.

• A car travels 60 km in 1.5 h. Find average speed in km/h and m/s.

• A bus changes velocity from 5 m/s to 17 m/s in 4 s. Calculate acceleration.

• A body starts from rest and accelerates uniformly at 3 m/s² for 4 s. Find final velocity and displacement.

• A runner completes one 200 m lap and returns to the starting point in 50 s. Find distance, displacement, average speed and average velocity.

19. ANSWER CHECK — SELECTED RESULTS

Question	Answer / check
C1	5 m/s
C2	40 km/h ≈ 11.11 m/s
C3	3 m/s²
C4	v = 12 m/s; s = 24 m
C5	Distance = 200 m; displacement = 0 m; average speed = 4 m/s; average velocity = 0 m/s

20. CHAPTER TAKEAWAYS

Motion is a change in position with time relative to a reference point. Distance and displacement describe how position changes; speed and velocity describe the rate of motion; acceleration describes the rate of change of velocity. Graphs and equations help us analyse motion and solve numerical problems.

Study smart • Think scientifically • Keep learning
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